

Simplified MVP Requirements

Waste Journey Tracking Prototype | Developer Brief

Purpose

Build a small functional web prototype demonstrating how a waste batch can be tracked through its main journey stages. This is an academic research/demo MVP, not a full municipal-scale production system.

1. Essential features (in scope)

Feature	Minimum expected functionality
Batch creation	Admin creates a waste batch; the system assigns a unique Batch ID.
QR code	Generate a QR code linked to the batch tracking page. Download/print support is optional if it adds cost.
Four stages	Record updates for Collection, Transportation, RTS/Transfer, and Processing.
Stage update	Each update records stage, date/time, manually entered location, and a short note/status. Quantity is optional if feasible.
Role-based access	Basic login with six roles: Admin, Collection, Transportation, RTS, Processing, and Head Officer. Stage users can update only their relevant stage; Head Officer has view-only access to the full journey; Admin can manage batches and users.
Tracking timeline	Display recorded stages in chronological order on the batch page.
Batch search	Open a batch by Batch ID. Scanning its QR opens the same tracking view.
Simple dashboard	Show basic counts such as total batches and batches by current stage/status; no advanced analytics.
CSV export	Export batch and stage tracking records to a CSV file for research and basic data analysis.
Web access	Simple responsive interface for laptop and mobile browser. Shareable demo link, subject to hosting arrangements.

2. Excluded for this MVP

- Live GPS, live maps, route optimization, or automatic vehicle tracking
- Blockchain network/integration (the prototype demonstrates digital traceability only)
- IoT devices, sensors, or automatic data capture
- Advanced permission rules beyond the six basic roles described above
- Master data management for routes, vehicles, facilities, or users
- Advanced analytics, reports, charts, and Excel export
- Email/SMS/OTP notifications, push notifications, or real-time updates
- Native mobile app, offline mode, or enterprise-scale deployment
- Complex approval workflows, audit trails, and advanced correction/version history
- Paid third-party APIs or integrations

3. Suggested workflow

Admin creates a batch and assigns it to the relevant stage users → system generates Batch ID and QR → authorized stage users add updates as the batch moves through Collection → Transportation → RTS/Transfer → Processing → Head Officer can view the journey → Admin can export records as CSV.